

Saatwik Sairaam Vasamsetti

Full-Stack Software Engineer · Data & Analytics

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PROFESSIONAL SUMMARY

Full-stack engineer who ships complete systems and proves they work. Sole author of an emergency-response platform — a 64-route TypeScript API on PostgreSQL/PostGIS, a Kotlin Compose Android client and a YOLO11 vision pipeline — whose API alone passes 636 assertions across six suites, 74 of them security attacks, plus 66 Android and 39 pipeline tests. Strong in data too: scikit-learn back-testing, statistical validation and BI, with 12 certifications across AWS, Google, SAP and Oracle. Available immediately for software-engineering and data internships.

EDUCATION

VIT-AP University, Amaravati

Integrated M.Tech — Computer Science & Engineering (in collaboration with Virtusa)

2024 – 2029

CGPA: 7.55 / 10

SR Junior College — Intermediate (Class XII, MPC)

2022 – 2024 · 94.8%

DAV Public School — Secondary School (Class X)

81.17%

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Kotlin, SQL, Java, HTML, CSS

Backend & Data Stores: Node.js, Fastify, Express, FastAPI, REST design, JWT/RBAC auth, PostgreSQL 16 + PostGIS, MySQL, Drizzle, Prisma, schema & index design, transactional concurrency

Frontend: React 19, Next.js 15, Angular, Tailwind CSS, Jetpack Compose, progressive web apps, dependency-free canvas UI, accessibility

Data, ML & BI: pandas, NumPy, scikit-learn, Ultralytics YOLO11, ONNX, EDA, walk-forward back-testing, conformal prediction, Monte Carlo, Power BI, Tableau, Matplotlib, Excel dashboards

Cloud, Tooling & QA: AWS, Oracle Cloud (OCI), Docker & Compose, Git, GitHub Actions CI, Gradle, Vitest, ESLint

PROJECTS

RoadAssist Bharat — Offline-First Emergency Mobility Platform

Aug – Sep 2026

Stack: TypeScript, Fastify, PostgreSQL 16 + PostGIS, Drizzle, Kotlin, Jetpack Compose, Python, YOLO11, Docker, GitHub Actions

- Sole author of a 64-route Fastify/TypeScript API over a 56-table PostGIS schema, plus a Kotlin Compose Android client and four browser surfaces — 117 commits.
- Verified the API with 636 assertions across six suites, zero failures, including 75 concurrency tests and 74 security attacks that must each be refused.
- Trained four YOLO11 variants on RDD2022 across four countries — best measured mAP50 0.471 on held-out validation, plus a separate nano ONNX tier at ~48 ms/image on CPU for Pi-class hardware.

Trade Research — Verifiable Research & Execution Platform

Aug – Sep 2026

Stack: Python, scikit-learn, pandas, FastAPI, PostgreSQL, React, MetaTrader 5, PyInstaller

- Built a leakage-free walk-forward back-test of six scikit-learn models against naive and drift baselines, with conformal intervals, Monte Carlo paths and a cost-aware trading simulation.
- Made the numbers checkable: indicators computed in Python, financials from SEC EDGAR XBRL, and a verifier that traces every figure in a generated note back to the data — then reported the honest negative result.

SkillForge — Entrepreneurship Enablement Platform

2026

Stack: TypeScript, Next.js 15, React 19, Express, Prisma, PostgreSQL 16, Tailwind CSS, Docker

- Architected a Next.js 15 / React 19 front end over an Express + Prisma API on PostgreSQL 16, containerised with Docker Compose.
- Built a recommendation engine scoring 150+ business ideas against a user's skills, budget and time, behind JWT auth with rotating refresh tokens, OTP verification and RBAC.

MachineSense AI — Explainable Industrial Intelligence

Smart India Hackathon 2026

Stack: Python, FastAPI, JavaScript, Canvas

- Built a FastAPI predictive-maintenance platform: weighted 0-100 health scoring, MAD z-score anomaly detection, remaining-useful-life forecasting and fleet energy optimisation.
- Made every score explainable by sensor with plain-language root causes, and added a digital twin that simulates a maintenance action before anyone touches the machine.

EXPERIENCE & LEADERSHIP

Campus Ambassador — SmartED

Jun 2026

- Led campus outreach for SmartED's initiatives; formally recognised for leadership, dedication and the connections built across the student community.

CERTIFICATIONS & RECOGNITION

- AWS Training & Certification:** Introduction to AWS · Cloud Practitioner (CLF-C02) Domain 1 · Data Engineer – Associate (DEA-C01) Exam Prep · **Oracle:** OCI Foundations Associate
- Google:** Analytics Certification (ID 183723043) · AI Essentials · **SAP Learning:** SAP Analytics Cloud · SAP Data Architect · **freeCodeCamp:** Responsive Web Design
- Forge Job Simulations:** Deloitte Data Analytics · Quantum Data Analytics · GenAI-Powered Data Analytics · **Recognition:** Campus Ambassador, SmartED (2026) · Participant, HackDevengers 1.0 (2026)